Codimension-4 degenerations of nilpotent orbit closures and type AIII symmetric pairs

Mengwei Hu

Johns Hopkins University

AMS 2026 Fall Southeastern Sectional Meeting
Geometric and homological methods in representation theory
October 10–11, 2026

Overview:
Nilpotent orbits in $\mathfrak{sl}_n(\mathbb{C})$ are indexed by partitions $\lambda \vdash n$.
Let S_μ be the Slodowy slice to a point $e \in \mathbb{O}_\mu \subset \overline{\mathbb{O}}_\lambda \setminus \mathbb{O}_\lambda$.
Consider the **nilpotent Slodowy slice**

$$S_\mu \cap \overline{\mathbb{O}}_\lambda.$$

*$\mathrm{Sing}(e, S_\mu \cap \overline{\mathbb{O}}_\lambda) \sim \mathrm{Sing}(e, \overline{\mathbb{O}}_\lambda)$ is smooth equivalence [KP81].
Let $\theta\colon \mathfrak{sl}_n(\mathbb{C}) \to \mathfrak{sl}_n(\mathbb{C})$ be a Lie algebra anti-involution.
E.g. $\theta\colon M \mapsto M^t$ or $\theta = -\mathrm{Ad}(\mathrm{diag}(I_p, -I_q))$.

Goal: Classify the singularity $(S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta$, when θ arising from an AIII symmetric pair and $\mathrm{codim}_{\mathbb{C}}(\mathbb{O}_\mu, \overline{\mathbb{O}}_\lambda) = 4$.
*$(S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta$ is a Lagrangian subvariety in $S_\mu \cap \overline{\mathbb{O}}_\lambda \rightsquigarrow \dim 2$.

Motivation: Classify irreducible Harish-Chandra $(\mathfrak{g}, K)$-modules in geometric terms. Roughly speaking, Supp(HC modules) $\approx (S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta$.

1. Review of Kraft–Procesi minimal degenerations
2. Codimension-4 degenerations
3. Involutions of quiver varieties and symmetric pairs

Minimal degenerations

Nilpotent $\mathrm{SL}_n(\mathbb{C})$-orbits in $\mathfrak{sl}_n(\mathbb{C})$ are indexed by partitions $\lambda \vdash n$.

$$\mu < \lambda \quad \Longleftrightarrow \quad \mathbb{O}_\mu \subset \overline{\mathbb{O}}_\lambda \setminus \mathbb{O}_\lambda.$$

Minimal degeneration

There is no partition ν with $\mu < \nu < \lambda$.

Kraft–Procesi '81 + Maffei '05

If $\mu < \lambda$ is a minimal degeneration, then the singularity of $S_\mu \cap \overline{\mathbb{O}}_\lambda$ is isomorphic to the singularity of type A_r or a_r, via simultaneous row and column reductions.

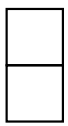
Hasse diagram in $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$

$\mathfrak{sl}_2$



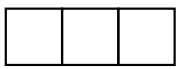
(2)

$xy - z^2 = 0 \mid A_1 = a_1$



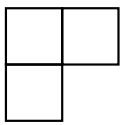
$(1,1)$

$\mathfrak{sl}_3$



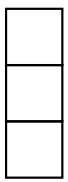
(3)

$xy - z^3 = 0 \mid A_2$



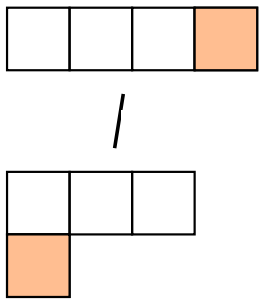
$(2,1)$

$\mid a_2$



$(1,1,1)$

A_r/a_r singularity and row/column removal



$(r+1)$

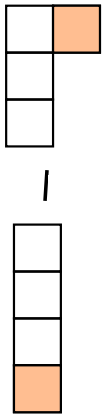
$|A_r$

$(r,1)$

$(2,1^{r-1})$

$|a_r$

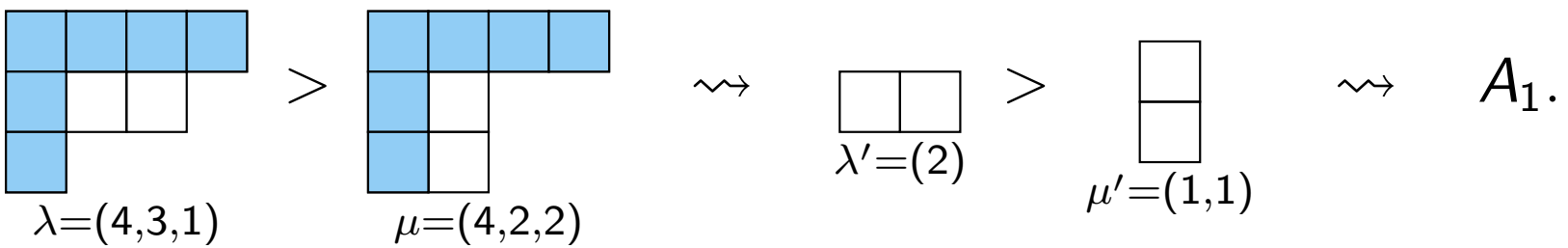
(1^{r+1})



$A_r = \{xy - z^{r+1} = 0\} \subset \mathbb{C}^3$ $\qquad$ $a_r = \overline{\mathbb{O}}_{\min}(\mathfrak{sl}_{r+1}) \leftarrow T^*\mathbb{P}^r$

Kleinian singularity $\rightsquigarrow$ dim 2 $\qquad$ dim $a_r = 2r$

Kraft–Procesi's removal of common row and columns (blue boxes)



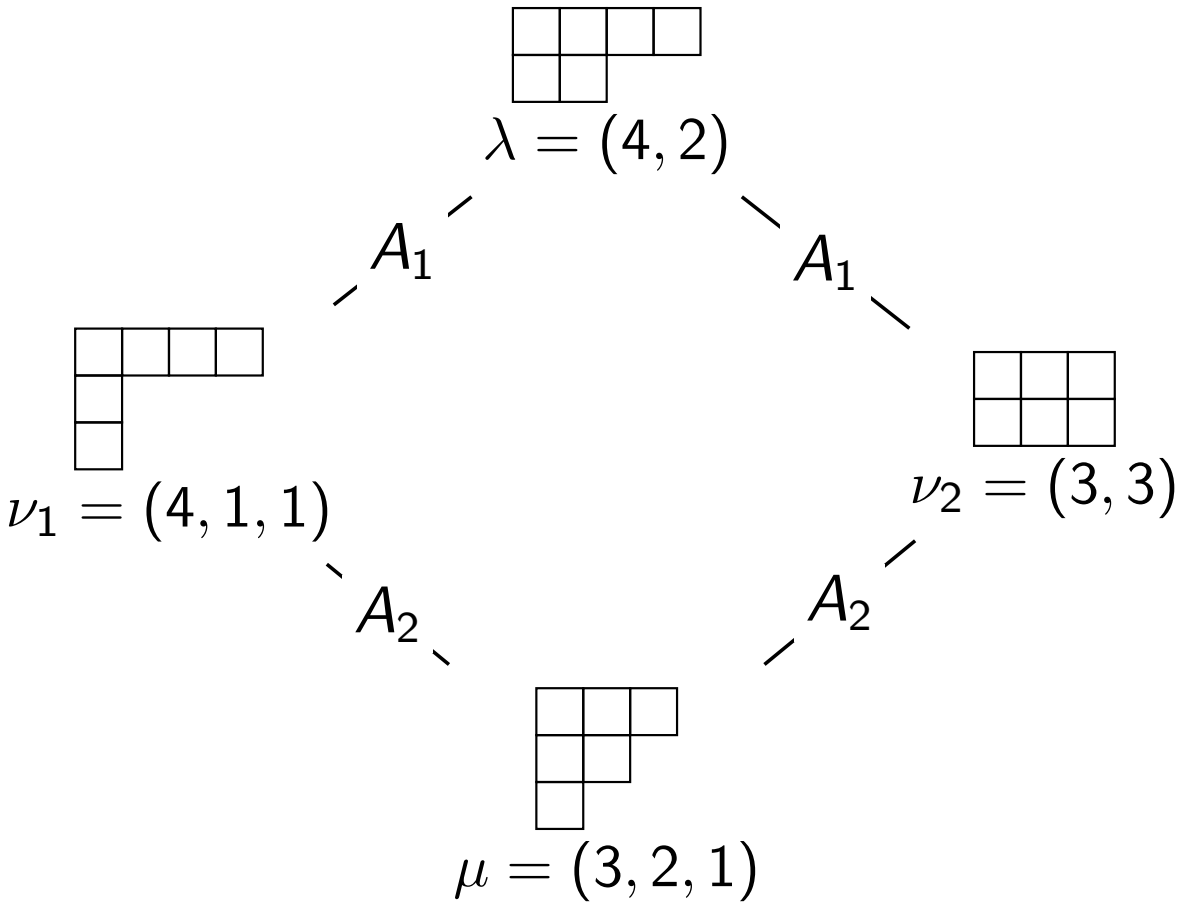
$\rightsquigarrow \quad A_1.$

Codim-4 degenerations: $a_2 = \overline{\mathbb{O}}_{\min}(\mathfrak{sl}_3)$ or two steps of A_r's.

Non-minimal degenerations

What happens when there are intermediate orbits ν occur?

$$\lambda = (4,2), \qquad \mu = (3,2,1), \qquad \dim S_\mu \cap \overline{\mathbb{O}}_\lambda = 4.$$



What is the singularity of the entire 4d variety $S_\mu \cap \overline{\mathbb{O}}_\lambda$?

Combinatorial classification of codimension-4 degenerations

A combinatorial classification

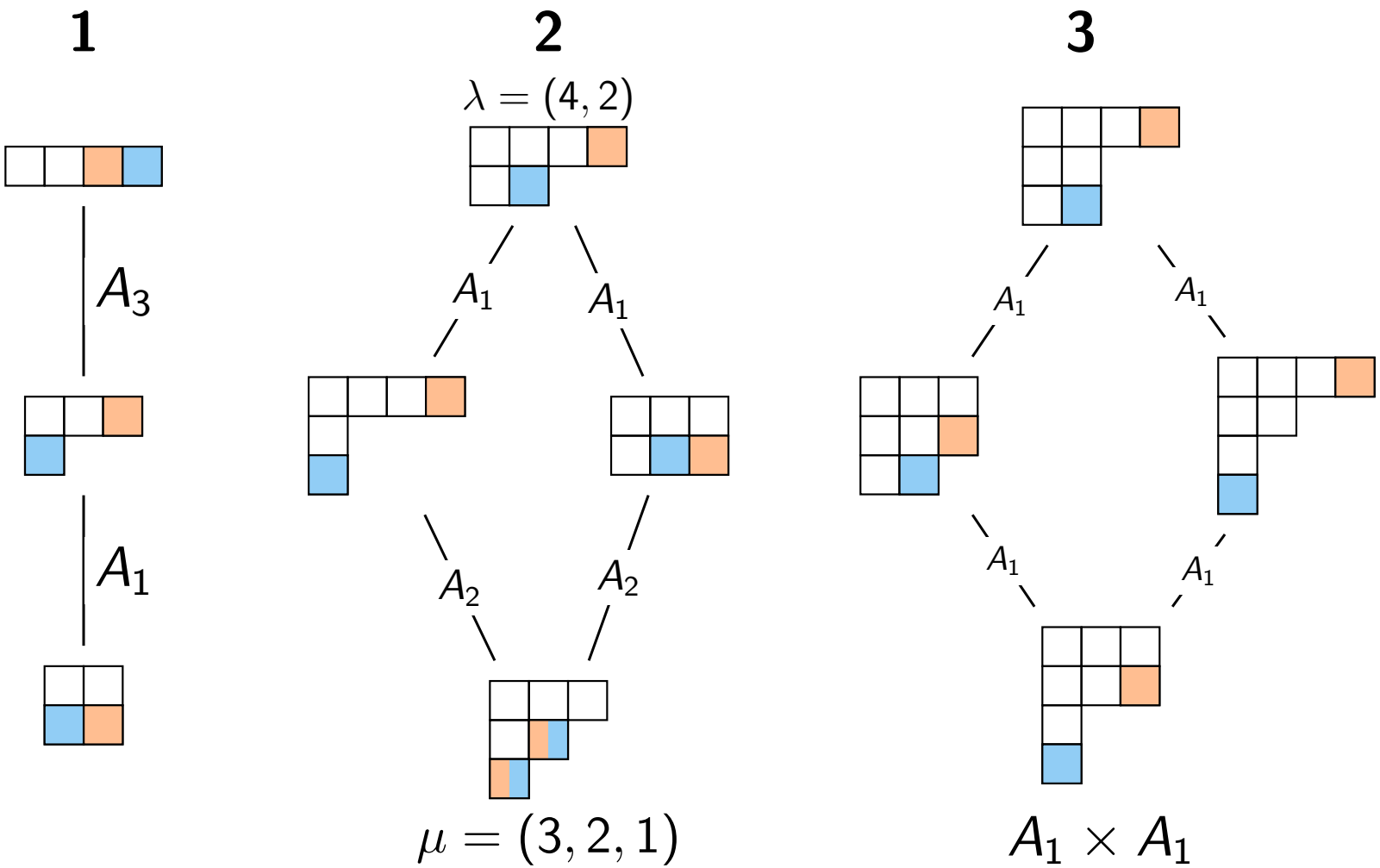
For $\mu < \lambda$ with $\dim S_\mu \cap \overline{\mathbb{O}}_\lambda = 4$, after simultaneous row and column removals, the pair $\mu < \lambda$ is one of the following cases:

1. $(n) > (n-2, 2)$
2. $(m, n) > (m-1, n, 1)$
3. $(m, k, n) > (m-1, k+1, n-1, 1)$

Remark:

- In Case 1, $\overline{\mathbb{O}}_{(n)} = \mathcal{N}$, the nilpotent cone.
- In Case 3, $S_\mu \cap \overline{\mathbb{O}}_\lambda \simeq A_{m-k-1} \times A_{n-1}$.
- We focus on Case 2 today! Note that Case 2 includes the minimal nilpotent orbit closure a_2 $(2,1) > (1,1,1)$ in $\mathfrak{sl}_3(\mathbb{C})$.

Three examples and their intermediate orbits



Ambient variety: $S_\mu \cap \overline{\mathbb{O}}_\lambda$ in Case 2

Proposition (H., in preparation)

In Case 2, we have

$$\mathbb{C}[S_\mu \cap \overline{\mathbb{O}}_\lambda] \simeq \frac{\mathbb{C}[x_1, x_2, x_3, y_1, y_2, y_3, z_1, z_2]}{I_2(M)},$$

where M is a 3×3 matrix whose entries are monomials of the generators, and $I_2(M)$ is the ideal generated by its 2×2 minors.

Examples of M:

$(4,2) > (3,2,1)$

$$\begin{pmatrix} z_1^2 & x_1 & x_3 \\ y_1 & -(z_1+z_2) & x_2 \\ y_3 & y_2 & z_2^2 \end{pmatrix}$$

$\overline{\mathbb{O}}_{\min}(\mathfrak{sl}_3) = a_2$

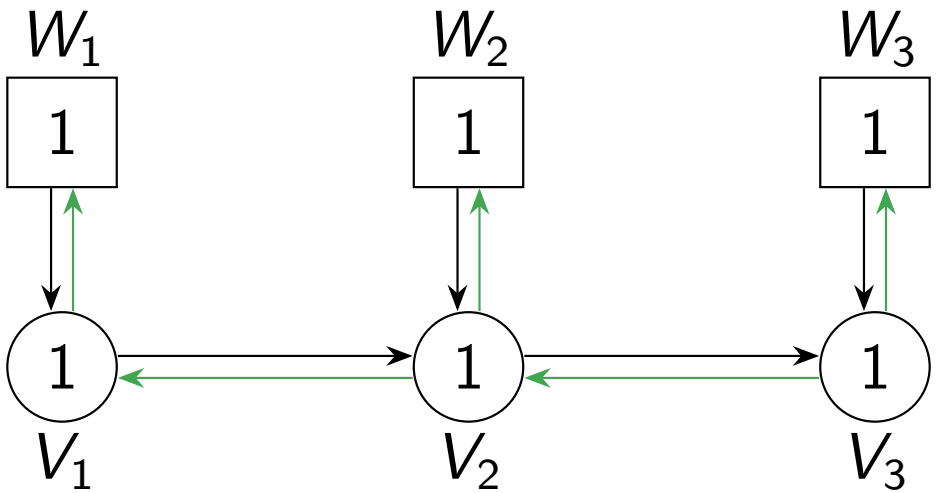
$$\begin{pmatrix} z_1 & x_1 & x_3 \\ y_1 & -(z_1+z_2) & x_2 \\ y_3 & y_2 & z_2 \end{pmatrix}$$

Proof technique: realize $S_\mu \cap \overline{\mathbb{O}}_\lambda$ as a quiver variety

The Nakajima–Maffei isomorphism gives

$$\Phi : \mathfrak{M}_0(\mathbf{v}, \mathbf{w}) \xrightarrow{\sim} S_\mu \cap \overline{\mathbb{O}}_\lambda.$$

For $(4, 2) > (3, 2, 1)$, take $\mathbf{v} = \mathbf{w} = (1, 1, 1)$:



The generators x_i, y_i, z_i are given by linear functions between the squares (framing). Relations come from associated moment map.

$z_1 : W_1 \to W_1, \qquad z_2 : W_3 \to W_3$

$x_1 : W_1 \to W_2, \qquad x_2 : W_2 \to W_3, \qquad x_3 : W_1 \to W_3$

Symmetric pairs and involutions of quiver varieties

For an involution $\sigma\colon \mathfrak{g} \to \mathfrak{g}$, set $\theta := -\sigma$. Write

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \qquad \mathfrak{k} = \mathfrak{g}^{\sigma}, \quad \mathfrak{p} = \mathfrak{g}^{\theta}.$$

Type	**Symmetric pair** $(\mathfrak{g}, \mathfrak{k})$
AI	$(\mathfrak{sl}_n, \mathfrak{so}_n)$
AII	$(\mathfrak{sl}_n, \mathfrak{sp}_n)$, n even
AIII	$(\mathfrak{sl}_n, \mathfrak{s}(\mathfrak{gl}_p \oplus \mathfrak{gl}_q))$, $p+q=n$

Type AIII is **inner**: $\sigma = \mathrm{Ad}\,\mathrm{diag}(I_p, -I_q)$.
Recall the Nakajima–Maffei isomorphism $\Phi : \mathfrak{M}_0(\mathbf{v}, \mathbf{w}) \xrightarrow{\sim} S_\mu \cap \overline{\mathbb{O}}_\lambda$,
Y. Li (2019) constructs an involution of quiver varieties θ_Q by *taking transposition with respect to bilinear forms*, which satisfies

$$\Phi \circ \theta_Q = \theta \circ \Phi,$$

$$\mathfrak{M}_0(\mathbf{v}, \mathbf{w})^{\theta_Q} \simeq (S_\mu \cap \overline{\mathbb{O}}_\lambda)^{\theta} = S_\mu \cap \overline{\mathbb{O}}_\lambda \cap \mathfrak{p}.$$

when θ arises from a symmetric pair of types **AI and AII**.

AIII involution and the Case 2 fixed-point locus theorem

In work in preparation, I construct an involution of quiver varieties by *assigning signatures to the framings*, together with appropriate sign changes of the quiver maps, realizing the **AIII** symmetric pair.

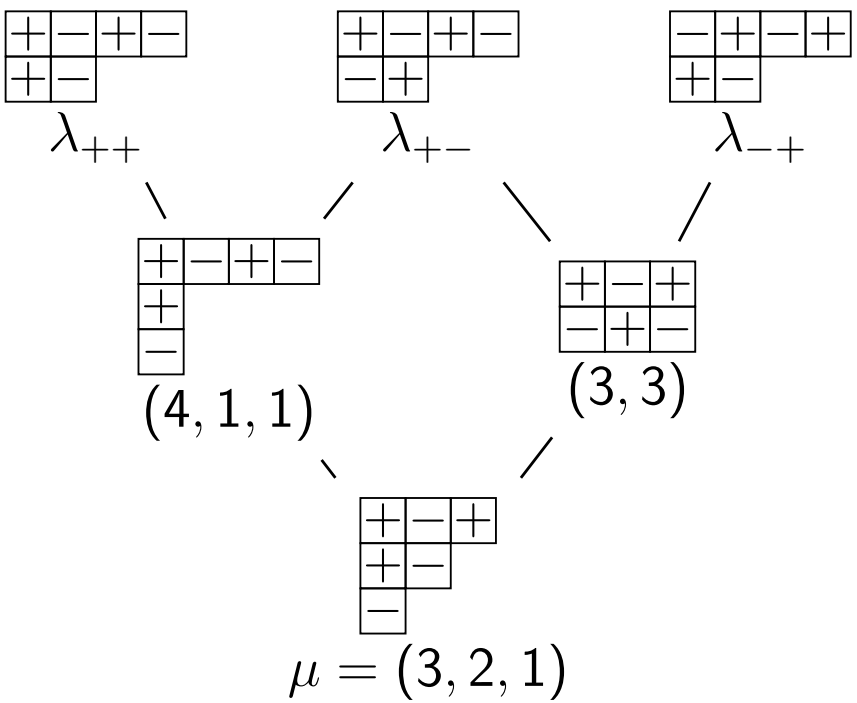
Theorem (H., in preparation)

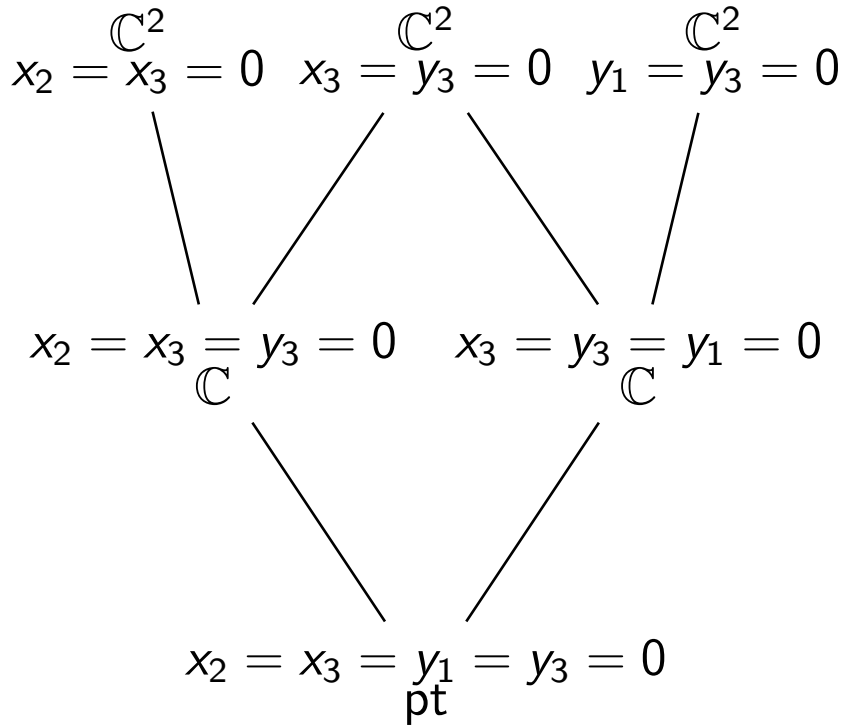
In Case 2, the fixed-point locus $(S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta$ is a union of copies of $\mathbb{C}^2$, meeting along lines or at points.

Remark: The AIII involution of quiver varieties construction work for any dimension and framing vectors $\mathbf{v}, \mathbf{w}$. For the fixed-point loci we care about, we just applied it to the 4d case.

The signed Hasse diagram: $(4,2) > (3,2,1)$

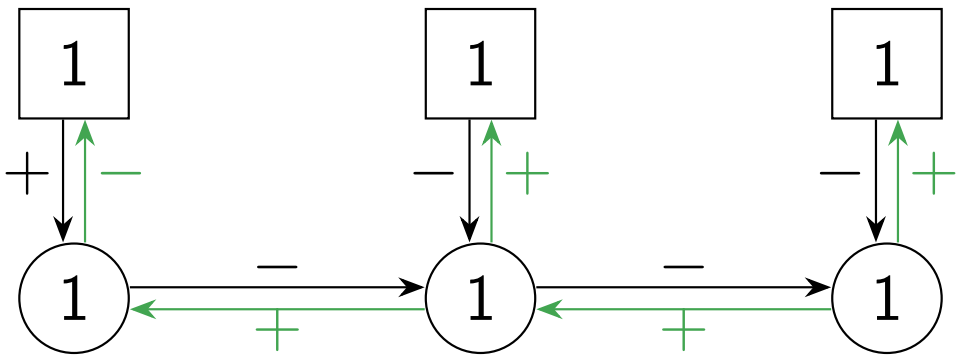
$$(S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta \simeq \operatorname{Spec} \mathbb{C}[x_2, x_3, y_2, y_3]/(x_2y_3, x_3y_3, x_3y_2)$$





Example of involution of quiver varieties

For $(4,2) > (3,2,1)$ with the above signed μ, the AIII involution of quiver variety is given by



We have $\theta(z_i) = -z_i; \theta(x_1) = -x_1, \theta(y_2) = -y_2$, resulting in that

$$(S_\mu \cap \overline{\mathbb{O}}_\lambda)^\theta \simeq \operatorname{Spec} \mathbb{C}[x_2, x_3, y_1, y_3]/(x_2y_3,\ x_3y_3,\ x_3y_1).$$

Thank you!